PARTNER PRODUCT: MKBANK, CREDIT MONITORING DEPARTMENT

Maqsad

Remotely verifies that a subsidised loan was spent on its stated purpose, using the money trail, photos and invoices. A bank officer makes the decision.

FinTech / AI in Finance · WIUT Hackathon 2026

PROBLEM

Part of subsidised lending never reaches its purpose

68 bn

UZS of subsidised loans misused in 2,500 cases found by regional inspections

~3 trn

UZS of family-business programme loans still not repaid — misuse is one of the drivers

Monitoring is the bank's duty. Today an officer visits each borrower in person — and borrowers are many and scattered

Sources: anhor.uz (meeting with bank heads); yuz.uz; Presidential Resolution PQ-5041

SOLUTION

Three kinds of evidence, one queue, a human decision

Money trail — the core signal

Cashed out on day one? Supplier paid? The bank's own data

Asset photo

Matches the purpose? Not a duplicate? Place and time

Invoice

Does the supplier's business code (OKED) fit the purpose?

Risk queue

- Red — check first
- Yellow — watch
- Green — remote check is enough

The **officer** approves, rejects or schedules a visit

LIVE DEMO

Try it yourself — scan with your phone

1 • Risk queue

Open a red loan — see why it is red

2 • Photo check

Pick the sewing-machine loan, photograph any object

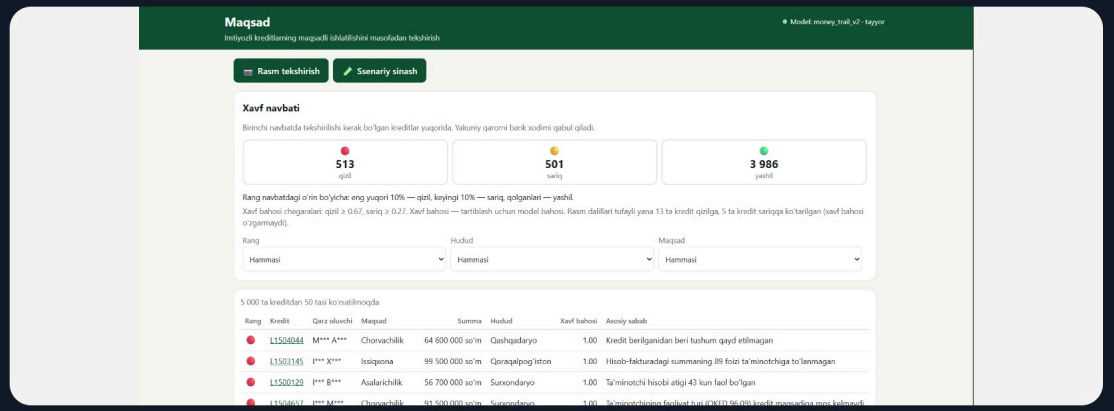
3 • Scenario

Enter your own money flows, get a risk score

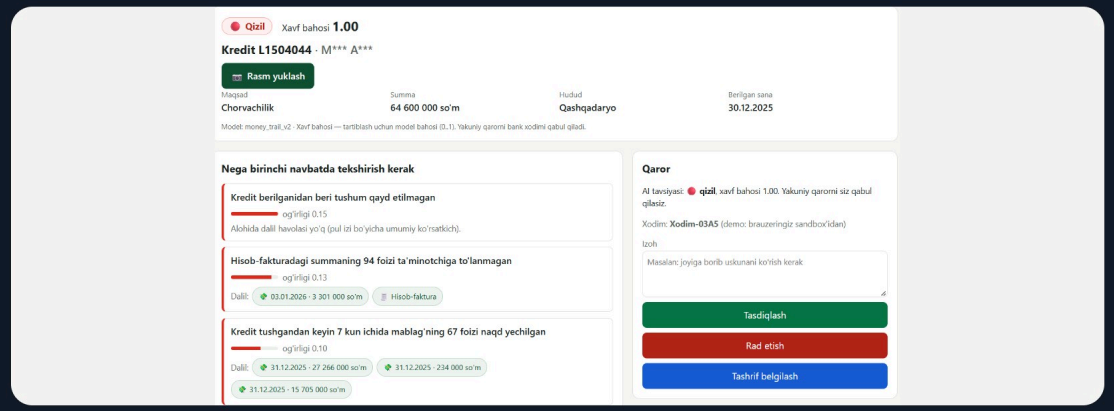
maqsad-production-d704.up.railway.app · every visitor gets a private sandbox — your results are visible only to you

DEMO · BACKUP

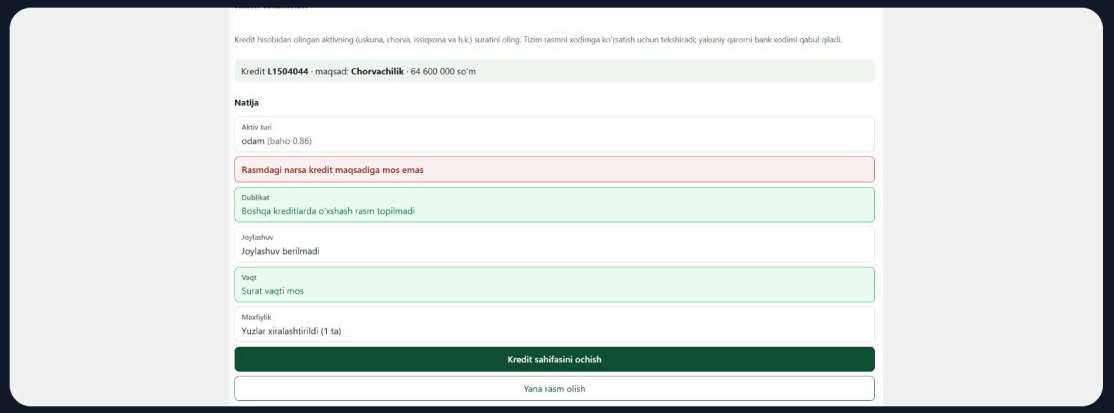
Four screens the officer sees



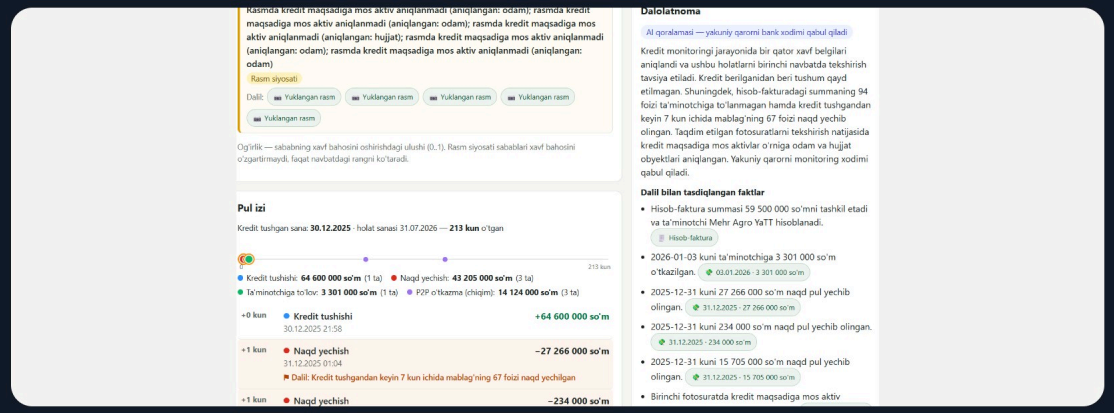
1 · Risk queue — red loans come first



2 · Why red: each reason linked to evidence



3 · Photo check: does not match the purpose



4 · Money trail and draft report

TECHNOLOGY

Models we built, quality we measured

78.2%

of fraud cases caught by **Maqsad** while visiting only 20% of loans

27.4%

caught by a simple rule with the same 20% visit budget

94.8%

asset-type accuracy on 139 open-licence photos (CLIP + our linear probe)

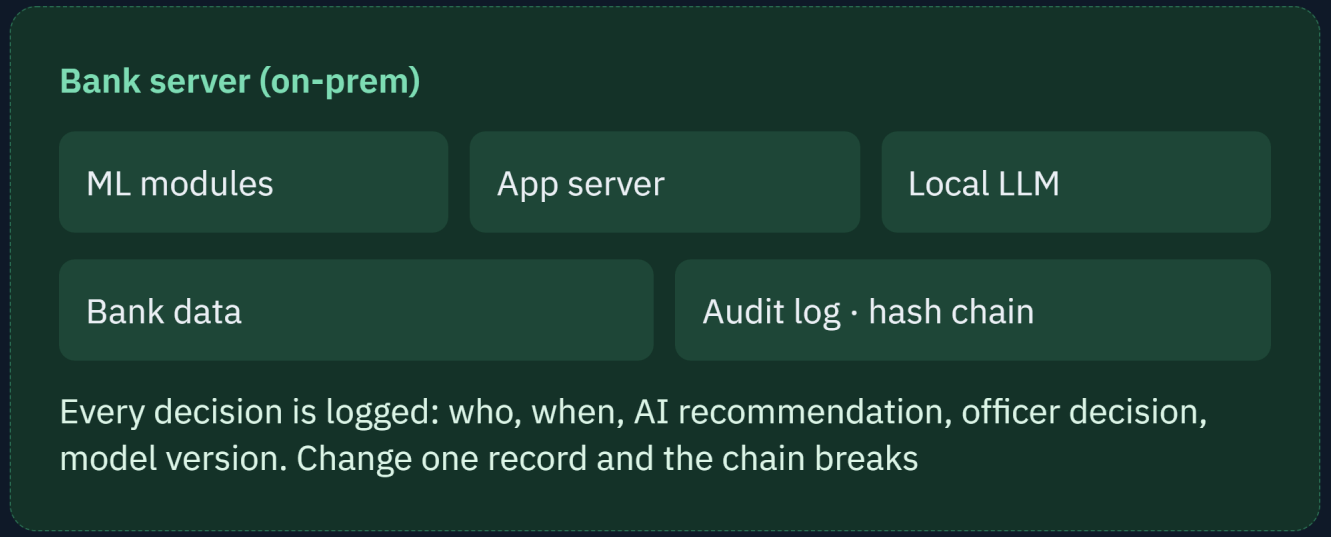
20% — random selection, same budget

Money trail: LightGBM with plain-language reasons · Photo: duplicate search, asset type, place and time, faces blurred · LLM only drafts the report; every fact is checked in code

78.2% and 27.4%: synthetic hold-out set, separate seed. 94.8%: real photos, cross-validation grouped by photographer, measured after removing personal data. Zero-shot CLIP alone: 89.9%.

DEPLOYMENT AND AUDIT

Customer data never leaves the bank



Measured on the live prototype	Value
Photo check, p50	363 ms
Queue (50 loans), p50	292 ms
Draft report, n=3	3.9–6.0 s
One draft report	$0.0034

Prototype LLM: Gemini API. Production: a local model — same code, only LLM_BASE_URL changes

BUILT-IN SAFEGUARDS

Known risks — and how the design handles them

Synthetic money trail → fraud patterns taken from real reported cases; month 1 of the pilot measures accuracy on the bank's historical loans

A photo can be staged → it never decides alone: duplicate search, place and time, and the money trail outweighs it

Model can miss a case → it only ranks the queue; an officer still decides and visits every red loan

LLM can make things up → every fact is checked against evidence in code; unverified facts are flagged, never hidden

Weakest case we measured: a fake invoice through a correctly registered supplier — 45% caught.
That is why the photo check and the officer stay in the loop.

ROI · ONE BRANCH, 1,540 LOANS

Remote checks pay for themselves fast

~3 mo

payback in the base case

~5 mo

even if half of the visits stay physical

~15k UZS

per remote check, versus **100–300k UZS** (assumed) for an on-site visit

Visit savings only; recovered losses are not included — measured in pilot month 1. Some inputs are assumptions; with all of them at worst case, ~34 months. Source: ROI model v3.3

PILOT PLAN

One branch, three months, measurable results

Month 1

Install and test

Install on the bank server, test on historical loans, measure how many cases today's method finds

Month 2

Run in parallel

Officers use the system alongside the current process; we tune it on their feedback

Month 3

Measure

Cases caught, visits saved and real accuracy — then the decision to scale

Existing AML tools look for money laundering, not for whether a loan was used for its purpose.
Proposed model: an annual licence per branch, priced below the visits it saves.

TEAM

Three people, three layers

Dilshodbek Tolibjonov

AI / ML

Synthetic portfolio, money-trail model, photo analysis, evaluation

Oybek Xushnazarov

Application

API, officer dashboard, camera, LLM report, audit, deployment

Shohijahon Shavkatov

Business and pitch

Problem evidence, ROI model, pilot plan

OUR ASK

A one-branch pilot with MKBANK

3 months · on the bank's server · the officer decides

78.2%
fraud caught visiting 20% of loans

~3 mo
payback in the base case

0
external APIs in production



Try the live site

Demo: maqsad-production-d704.up.railway.app · Code: github.com/dilshodbekio/maqsad · Synthetic money-trail data; real accuracy measured in the pilot